



Ahmed Aziz Ben Haj Hmida

Data Scientist (EPFL, MSc, GPA 5.3/6) combining deep ML expertise with a Minor in Management & Entrepreneurship. Specialized in designing AI systems that solve high-stakes industrial and analytical problems — with a track record of cutting process times, enabling design reuse, and bridging technical teams with business objectives.

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Professional Experience

Rollomatic SA Le Landeron, Switzerland
Master's Thesis: Strategic Retrieval System for CNC Tools (Grade 6/6)
Aug 2025 – Mar 2026

- Engineered an end-to-end similarity search system deployed into active R&D workflows, reducing search latency by 96% and enabling tool design retrieval in under 200ms instead of hours.
- Improved retrieval accuracy by 11%, directly enabling design reuse, accelerating tool development cycles, and reducing manufacturing R&D costs at scale.

Supervisors: Prof. Jean-Philippe Thiran - jean-philippe.thiran@epfl.ch, Emna Maghrebi - e.maghrebi@rollomatic.ch

Rollomatic SA Le Landeron, Switzerland
Internship: Data-Driven Optimization for Laser Machining
Feb 2025 – Aug 2025

- Mapped tool retrieval bottlenecks across laser machining units, identifying a 60%+ manual search overhead.
- Built a proof-of-concept hybrid search pipeline combining visual embeddings and metadata filtering, validating technical feasibility and securing stakeholder buy-in for the production system.
- Co-defined KPIs with R&D and operations teams, establishing the performance benchmarks (latency, recall@k) against which the production system was later evaluated.

Supervisor: Emna Maghrebi - e.maghrebi@rollomatic.ch

Technical Skills

Languages: Python (expert), SQL, Java, MATLAB
Business: Financial Modeling, Project Management, Risk Analysis, Power BI, Excel
AI & ML: PyTorch, Scikit-learn, RAG Pipelines, Agentic AI, LLMs (fine-tuning, RLHF), Computer Vision
Data & Search: FAISS, Milvus, ChromaDB, PostgreSQL, vector search, embeddings
MLOps: Docker, Kubernetes, MLflow, WB, DVC, CI/CD, FastAPI, AWS, GCP

Education

EPFL – École Polytechnique Fédérale de Lausanne Lausanne, Switzerland
M.Sc. in Data Science
Sep 2023 – Mar 2026
Minor in Management, Technology & Entrepreneurship (MTE)
GPA: 5.3/6

Relevant Coursework: Applied Data Analysis, Data Science for Business, Finance, Microeconomics, Project Management & Risk Analysis, Value Chain Management, Economic Growth & Sustainability

Technical University of Denmark (DTU) Lyngby, Denmark
3rd Year of Bachelor – Exchange
Aug 2022 – Jul 2023
GPA: 5.35/6

EPFL – École Polytechnique Fédérale de Lausanne Lausanne, Switzerland
B.Sc. in Communication Systems
Sep 2020 – Jul 2023
GPA: 5.1/6

Pioneer School of Nabeul Nabeul, Tunisia
Baccalaureate in Mathematics
Sep 2018 – Jun 2019

Research Projects

Discovering promising new startups through evaluating teams' potentials 🔄 Sep 2023 – Dec 2023
EPFL ENTIC Lab

- Harnessed ML methods to derive speakers' MBTI personalities and emotions during entrepreneurial discussions.
- Built predictive approaches to identify promising teams with high startup success potential.

Supervisor: Prof. Davide Bavato

Unsupervised Reverse Knowledge Distillation and Model Merging 🔄 Aug 2024 – Feb 2025
EPFL LTS4 Lab

- Transferred knowledge from compact task-specific models into large-scale architectures.
- Applied model merging techniques to improve niche-task performance while preserving broad generative capabilities.

Supervisor: Prof. Pascal Frossard - pascal.frossard@epfl.ch, Ortal Senouf

Personal & Academic Projects

AI-Powered Food & Nutrition Assistant  May 2026 - Present

- Built a React frontend and Python backend with a LangChain ReAct agent for natural language meal logging, GPT-4o vision for photo-based meal analysis, and ChromaDB semantic search over 200k+ recipes enriched with USDA nutritional data.
- Tracked daily macro intake with personalised targets (Mifflin–St Jeor), automated email digests via an MCP tool, and deployed to AWS with a GitHub Actions CI/CD pipeline and CloudWatch monitoring.

LLM Agent–Driven Car Search Assistant  Mar 2026 - May 2026

- Built a Streamlit frontend where users describe a car in chat and LangGraph with LangChain and Ollama fills a structured profile and queries PostgreSQL.
- Displayed ranked listing cars with match scores plus Plotly and Matplotlib charts, generated PDFs for single searches and for comparing two or three saved car profiles, and loaded the listings into Postgres.

LLM Teaching Assistant for MCQ Answering  Feb 2024 – Jun 2024

- Fine-tuned Qwen1.5 using SFT and Direct Preference Optimization with RAG and 4-bit quantization.
- Developed a chatbot interface and evaluation suite using LLM-as-a-judge frameworks and cross-validation.

Machine Learning Operations  Jan 2023 – Mar 2023

- Deployed scalable ML pipelines using PyTorch, Docker, Kubernetes, CI/CD, GCP, AWS, MLflow, and DVC.

RL for Motor Skill Transfer  Feb 2024 – Jun 2024
EPFL Alexander Mathis Lab

- Trained expert reinforcement learning agents on Myosuite baoding ball tasks.
- Distilled motor strategies into novice agents to improve learning efficiency and performance.

Coin Detector and Counter  Feb 2024 – Jun 2024

- Used segmentation techniques and deep learning models (CNNs, Vision Transformers, ResNet50) for accurate coin recognition.

Certificates

IBM RAG and Agentic AI – IBM	Apr 2026 - Jun 2026
DevOps and AI on AWS – AWS	May 2026 - Jun 2026
Structuring Machine Learning Projects – DeepLearning.AI	Sep 2023 - Oct 2023
Certificate in Artificial Intelligence – GOMYCODE	Jun 2023 – Sep 2023

Leadership & Engagement

EPFL Coaching – Mentored first-year students in academic strategy and study methods	Sep 2023 – Jun 2024
Rotary International – Member	Jul 2018 – Apr 2019
Tunisian Red Crescent – Volunteer	May 2018 – Apr 2019

Languages

Arabic/French: Native | English: C1 | German: A2